

Examples, q, p functions

$$q_i = \begin{cases} 0 & i \notin N(b) \\ \frac{1}{|N(b)|} & i \in N(b) \end{cases}$$

$$N(b) = \{i \in N : b_i = \max_{j \in N} b_j\}$$

$$p_i(i^*; b) = \begin{cases} 0 & i \neq i^* \\ \max_{j \neq i} b_j & i = i^* \end{cases} \quad \text{second price auction.}$$

$$q_i = \begin{cases} 0 & i \notin N(b) \\ \frac{1}{|N(b)|} & i \in N(b) \end{cases}$$

$$N(b) = \{i \in N : b_i = \max_{j \in N} b_j\}$$

$$\underline{N}(b) = \min \{i \in N : b_i = \max_{j \in N} b_j\} = \min N(b)$$

highest bidder always wins

$$q_i = \begin{cases} \frac{1}{n} & b = \bar{0} \\ \frac{b_i}{\sum_j b_j} & b \neq \bar{0} \end{cases}$$

Prob. of winning is monotonic with bid, but highest bid doesn't ensure winning

$$q_i(b_1, \dots, b_n) = \begin{cases} \frac{1}{n} & b = \bar{0} \\ \frac{b_i}{\sum_j b_j} & b > \bar{0} \end{cases}$$

Payment rules! Payoffs are $q_i(b_1, \dots, b_n) v_i(t) - \sum_{i^* \in N} q_{i^*}(b_1, \dots, b_n) p_i(i^*; b)$

First-price: $p_i(i^*; b) = \begin{cases} 0 & i \neq i^* \\ b_i & i = i^* \end{cases}$

Third price?

Second-price: $p_i(i^*; b) = \begin{cases} 0 & i \neq i^* \\ \max_{j \neq i} b_j & i = i^* \end{cases}$

let $s = \arg \max_{s \neq i} b_s$

$$B_{-i} = \{j : j \in \arg \max_{s \neq i} b_s\}$$

All pay $p_i(i^*; b) = b_i$ (constant across i^*)

All pay / second price: $p_i(i^*; b) = \begin{cases} b_i & \text{if } i \neq i^* \\ \max_{j \neq i} b_j & \text{if } i = i^* \end{cases}$

Entry fees? $p_i(i^*; b) = \begin{cases} f_i & \text{if } i \neq i^* \\ b_i + f_i & \text{if } i = i^* \end{cases}$
first-price

If $|B_{-i}| = 1 \rightarrow \max_{j \in N/B_{-i}} b_j$
If $|B_{-i}| \geq 2 \rightarrow \max_{j \neq i} b_j$

Theorem 5.4 (Proof)

Consider i with value v_i . Divide the bids available to her as:

- $b_i \in [0, v_i)$
- $b_i = v_i$
- $b_i \in (v_i, \infty)$

Denote the maximal bid put forward by bidders other than i

as $B_{-i} = \max_{j \neq i} b_j$

and the number of bidders who offered this bid as:

$$N_{-i} = \left| \left\{ k \neq i : b_k = \max_{j \neq i} b_j \right\} \right|$$

$$= \left| \operatorname{argmax}_{j \neq i} b_j \right|$$

The payoff $u_i(b; v_i)$ of buyer i as a function of b is

$$u_i(b; v_i) = \begin{cases} 0 & \text{if } b_i < B_{-i} \\ \frac{v_i - B_{-i}}{N_{-i} + 1} & \text{if } b_i = B_{-i} \\ v_i - B_{-i} & \text{if } b_i > B_{-i} \end{cases}$$

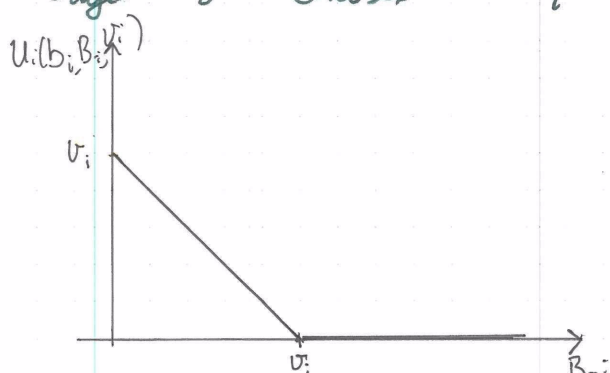
↑

this depends on b_{-i} only through B_{-i}

so we denote it by: $u_i(b_i, B_{-i}; v_i)$

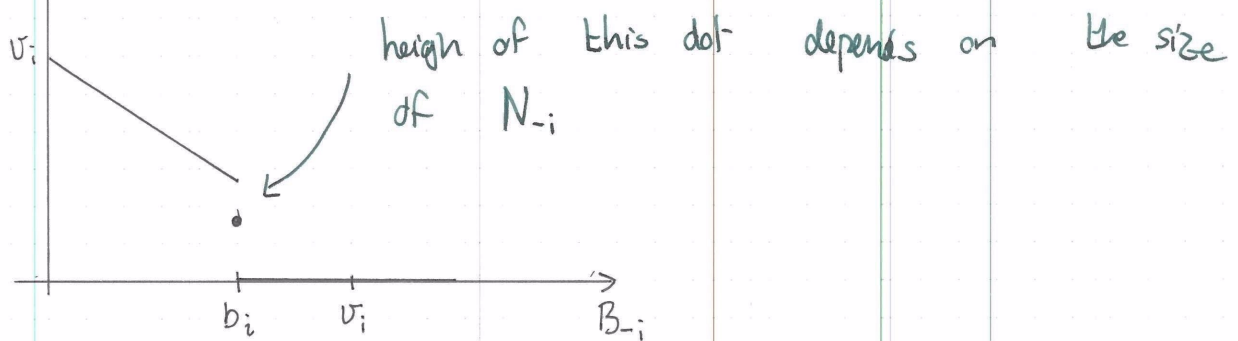
We consider the three cases above:

If buyer i chooses $b_i = v_i$ the payoff function looks like

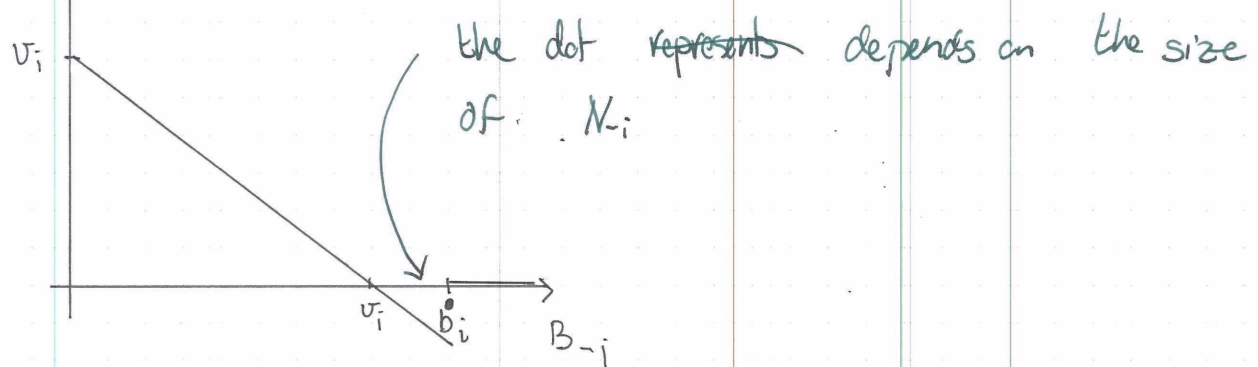


Bayesian Auctions

If buyer i chooses strategy b_i with $b_i < v_i$ her payoff function is $u_i(b_i < v_i, B_{-i}; v_i)$



If buyer i chooses strategy $b_i > v_i$ her payoff function looks like this



Note that:

- $u_i(b_i = v_i, B_{-i}; v_i) > u_i(b_i < v_i, B_{-i}; v_i)$ when $b_i \leq B_{-i} < v_i$ and equal otherwise
 - $u_i(b_i = v_i, B_{-i}; v_i) > u_i(b_i > v_i, B_{-i}; v_i)$ when $v_i < B_{-i} \leq b_i$ and equal otherwise
- $\Rightarrow b_i = v_i$ weakly dominates all other bids